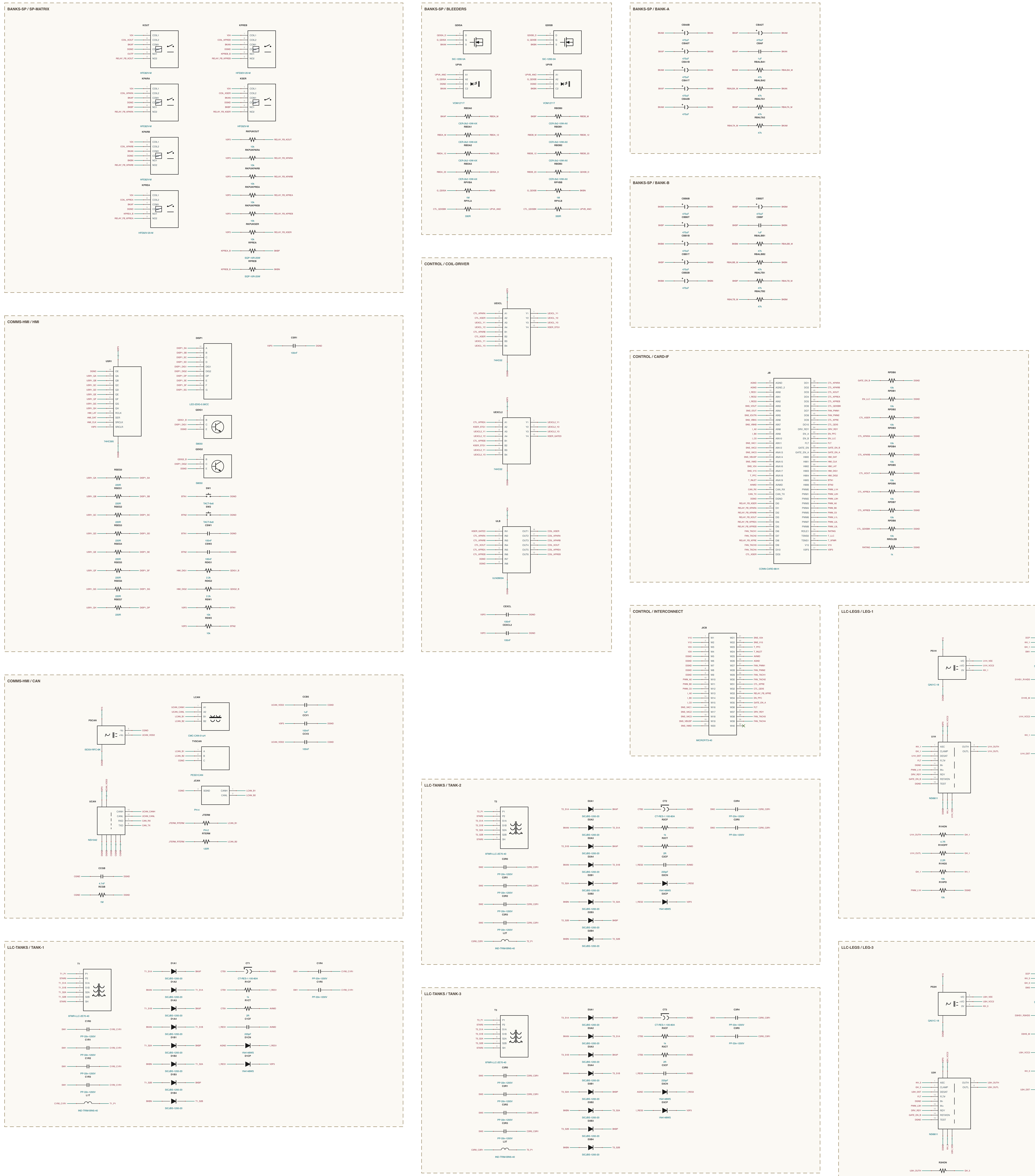


SHEET INDEX

40 kW DC-DC (upper) - 2 of 3

OUTPUT-SENSING - 6 sections

BANKS-SP - 4 sections

LLC-LEGS - 4 sections

CONTROL - 3 sections

LLC-TANKS - 3 sections

COMMS-HMI - 2 sections

rev D.4 - 22 sections - 319 components

NET NAMING

internal junctions are named for what they join

U<ref>_<PIN> node at that IC pin e.g. UIVOA_VINP

R<stem>_M midpoint of a series pair e.g. RBALTA_M

R<stem>_<nm> tap between R<stem>n/m e.g. RV1D_01

all others are explicit design nets

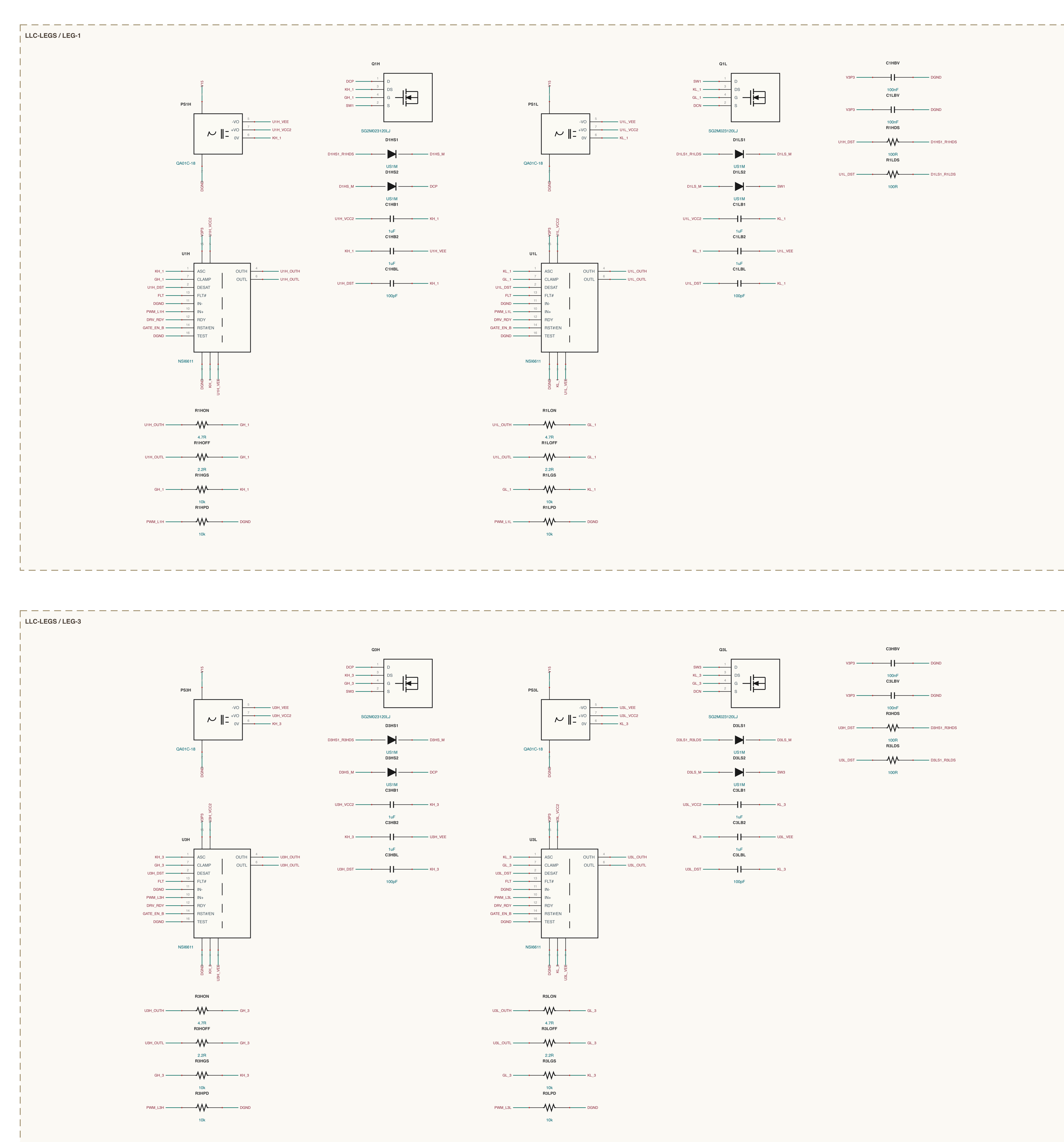
MAGNETICS CONSTRUCTION

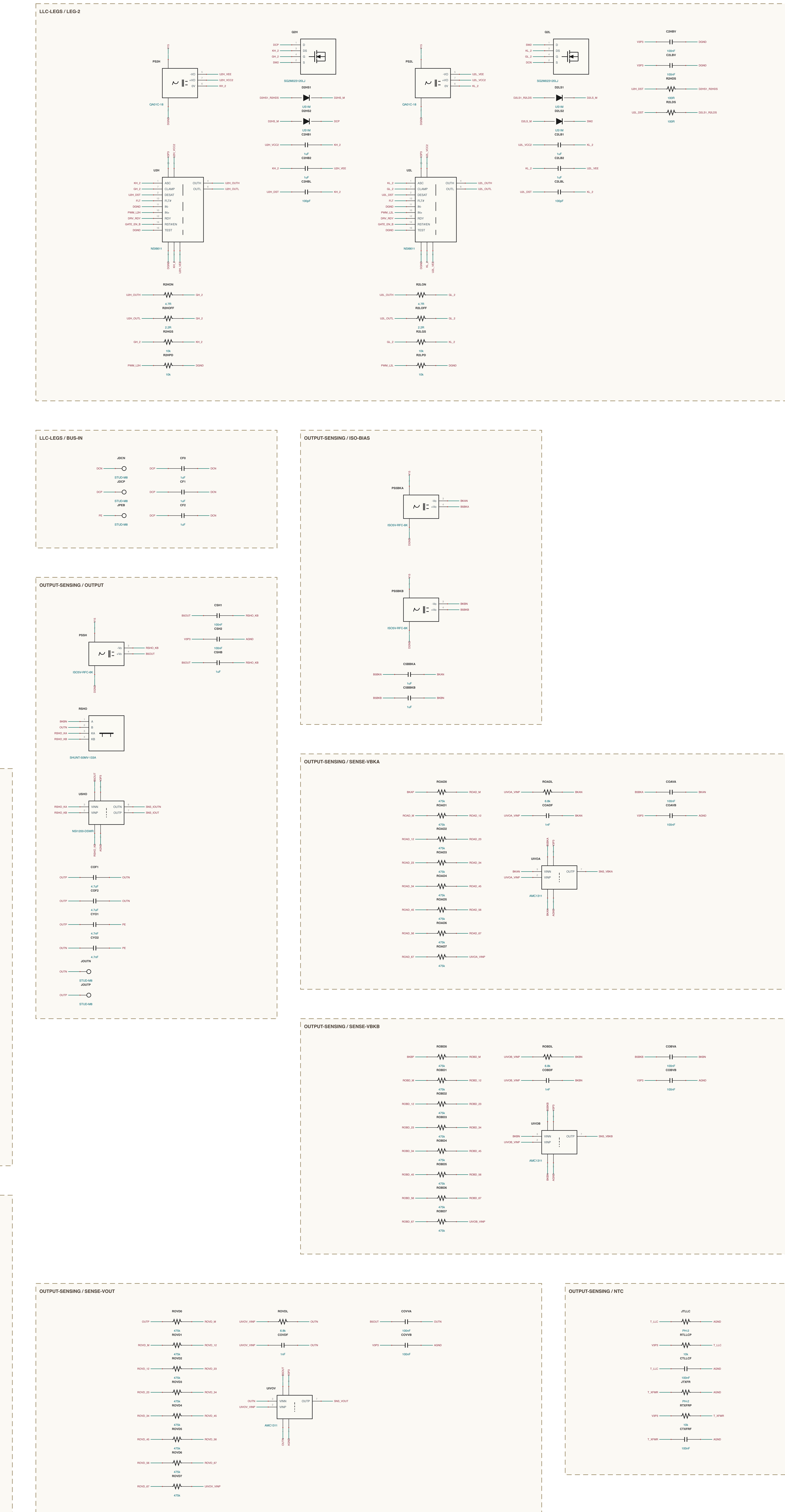
identity per docs/magnetics.md (turns, gap, litz, acceptance lines live there)

D3 T1-T3: 2x E70/33/32 per section, 9:9:9 (window-fill basis, D3-40)

D2 L1T-L3T: N=5, bins 3.2/3.5/3.8uH, litz 2000x0.1 (15.7mm2)

tank Cr/trim-bin values printed at the TANK sections





40 kW DCDC board 2of2 - 3-phase LLC (1x hot cells)

DC-Modules 40 kW - board DC-DC (upper), sheet 2 of 3

2026-09-06 · rev D.4